SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – COMPARITIVE ANALYSIS

Course Code:	ITA0312	**Course Title:**	Mobile computing
CO Assessed:	CO4	**Assessment:**	AT1 - COMPARITIVE ANALYSIS
Weightage:	30 % of CIA	**Total Marks:**	100
CO4 Statement:	*Identify the components and structure of mobile application for Android and Windows OS-based devices.*		
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Code of Conduct

I P.Ranjithkumar (192411119)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P.Ranjithkumar

Instructions

- Duration: 60 minutes
- Number of questions : 01

Faculty In-charge	Course Coordinator	Head of Department
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1.Introduction

Enterprise messaging applications allow employees to exchange confidential messages, documents, and work-related information. When an organization must support older corporate smartphones, the operating system must provide reliable security, suitable network connectivity, a practical application runtime, and efficient use of limited memory, processing power, and battery capacity.

BlackBerry OS and Symbian were prominent smartphone operating systems during the early enterprise-mobility era. BlackBerry developed a strong reputation for managed corporate messaging and centralized administration. Symbian was widely used across several manufacturers and provided platform-security mechanisms, but its capabilities and implementation could vary by device, edition, and vendor.

This assessment compares the two systems across four required areas: security architecture, networking support, application runtime, and resource management. The comparison refers to legacy BlackBerry OS and Symbian generations, not modern BlackBerry-branded Android devices.

2. Overview of the Operating Systems

Feature	BlackBerry OS	Symbian OS
Primary orientation	Strong focus on mobile communication, business email, device administration, and enterprise deployment.	General-purpose smartphone platform used by multiple handset manufacturers; strong emphasis on efficient operation on mobile hardware.
Typical enterprise strengths	Integrated messaging services, enterprise server integration, policy management, and security controls.	Broad device availability, multitasking, efficient resource use, and application development flexibility.
Security approach	Security features combined with enterprise infrastructure and administrator-controlled policies.	Platform-level protections such as application signing, capabilities, process isolation, and data caging, depending on version.
Main deployment concern	Legacy devices and services may be unsupported or incompatible with current enterprise infrastructure.	Device/vendor fragmentation and legacy software support can complicate consistent security management.

3. Security Architecture

3.1 BlackBerry OS

BlackBerry's traditional enterprise model combined device-level controls with enterprise messaging infrastructure. In deployments using BlackBerry Enterprise Server (BES), the server acted as a managed bridge between corporate messaging systems and enrolled devices. Depending on the device generation and configuration, organizations could apply IT policies, control supported services, authenticate users, and protect communications in transit. Encryption and authentication were important parts of the enterprise design.

This approach is particularly valuable when an organization needs consistent policy enforcement across a fleet of employee devices. A lost device can be handled through available administrative actions, such as disabling access or issuing a remote wipe command, when supported and correctly configured.

3.2 Symbian OS

Symbian OS versions 9 and later introduced a platform-security model that included capabilities, application signing, and data caging. Capabilities restrict applications from accessing sensitive system services unless they have the required permissions. Data caging limits access to protected areas of the file system, reducing the ability of one application to read another application's private data. Application signing helps establish the source and integrity of software, although signing alone does not guarantee that an application is safe.

These mechanisms provide meaningful protection, but the level of enterprise control depends on the Symbian version, handset vendor, installed applications, and any additional management software. Security settings and device behaviour may not be identical across all Symbian phones.

Security criterion	BlackBerry OS	Symbian OS	Comparative assessment
Authentication and access control	Enterprise deployments could combine device passwords, account controls, and administrator-defined IT policies.	Uses platform permissions and capabilities; authentication and enterprise access controls depend on device/software configuration.	BlackBerry is generally stronger for centrally administered enterprise access.
Application security	Application controls and enterprise policy tools could	Application signing, process separation, capability-based	Both offer protections;

	restrict software and services, depending on generation.	access control, and data caging on supported versions.	Symbian's capability model is technically significant.
Data protection	Enterprise configurations could protect messages and corporate information in transit and apply device-management policies.	Protected storage areas and application isolation help reduce unauthorized access; additional encryption depends on device and configuration.	BlackBerry's integrated enterprise messaging model is an advantage.
Central administration	BES-based environments offered centralized policy and messaging administration.	Often depended more on device-specific tools, vendor support, or third-party management solutions.	BlackBerry has the clearer enterprise-management advantage.
Lost or stolen device response	Available enterprise tools could support remote lock/wipe and account or service control, subject to configuration.	Remote actions generally depend on the device and management solution available.	BlackBerry is typically more straightforward in a supported managed deployment.
Security caveat	Legacy services, old encryption settings, and unsupported devices can create serious risks today.	Signing and capabilities do not prevent every malicious action or compensate for unpatched vulnerabilities.	Neither legacy platform should be assumed secure merely because it once had security features.

4. Networking Support

4.1 BlackBerry OS

Traditional BlackBerry smartphones were designed around always-connected messaging and push-style delivery. In enterprise deployments, BlackBerry infrastructure and BES could help route messages between the corporate messaging environment and employee devices. This architecture could simplify

administration and support predictable delivery of work email and messages. Actual connectivity depended on the handset, carrier, service plan, server version, and organizational configuration.

4.2 Symbian OS

Symbian supported common mobile networking technologies available on its devices, including cellular data and, on many models, Wi-Fi, Bluetooth, and TCP/IP-based networking. Applications could use network APIs to connect to web services or messaging servers. However, the exact protocols and connection options varied by handset and software version, so developers needed to account for differences between devices and networks.

Networking factor	BlackBerry OS	Symbian OS
Messaging model	Historically optimized for managed email and message delivery, including push-oriented enterprise workflows.	Messaging depended on native clients or third-party applications and their server connections.
Corporate integration	BES could integrate with corporate messaging systems and apply policies to managed devices.	Could connect to corporate servers using supported protocols, but often required application- and device-specific configuration.
Connectivity options	Supported cellular data and other device-dependent connections; enterprise traffic flow could be shaped by BlackBerry infrastructure.	Supported cellular data, Wi-Fi on many models, Bluetooth, and standard networking APIs, depending on handset.
Developer networking	Enterprise services could simplify messaging integration, but could create dependence on a particular legacy platform and infrastructure.	General networking APIs enabled flexible client-server applications, with compatibility testing required.
Key limitation	Legacy service availability and compatibility with modern systems must be verified.	Network features and reliability differ across manufacturers, models, and Symbian versions.

5. Application Runtime and Development

5.1 BlackBerry OS

Legacy BlackBerry devices commonly supported Java-based applications using the BlackBerry Java application environment, while later BlackBerry generations used different runtime models. This made it possible to develop messaging clients with device APIs and enterprise integration. A key benefit was the ability to build applications around the platform's communication and management ecosystem. The trade-off was dependence on specific APIs, device generations, and legacy development tools.

5.2 Symbian OS

Symbian applications could be developed using native C++ frameworks, and selected generations also supported Java ME and other application technologies. Native applications could use platform APIs and offer strong control over performance and device integration. However, development could be more complex because APIs, user-interface frameworks, signing requirements, and compatibility details differed across Symbian editions and device families.

Runtime criterion	BlackBerry OS	Symbian OS	Implication for messaging
Common legacy development model	BlackBerry Java environment on many classic devices; runtime varies by generation.	Native C++ and Java ME on supported devices; options vary by generation.	Confirm the exact handset and OS version before selecting a stack.
Development complexity	Enterprise APIs can simplify certain messaging workflows, but legacy toolchains can be difficult to maintain.	Native development offers flexibility but can require more platform-specific knowledge and testing.	BlackBerry may be simpler for a narrowly scoped legacy enterprise client.
Security controls	Can combine application controls with enterprise policies, depending on platform generation.	Signing and capability declarations regulate access to protected functionality on supported versions.	Both require secure coding, careful permissions, and trusted distribution.
Compatibility	Applications may be tied to a	Fragmentation across Symbian	Test on the actual

	particular BlackBerry generation or API set.	editions and manufacturers can increase testing effort.	fleet rather than relying on platform names alone.
Long-term maintainability	Legacy APIs and unsupported services may be difficult to integrate with current infrastructure.	Legacy toolchains and device fragmentation also create maintenance challenges.	Neither is ideal for a new long-lived application without a migration plan.

6. Resource Management

6.1 BlackBerry OS

BlackBerry smartphones were designed for mobile productivity, where background communication, message synchronization, and battery life were important. The platform's messaging-oriented behaviour could reduce the need for an application to repeatedly poll a server, depending on the service architecture. However, older devices had limited memory and processing capacity, so message storage, attachment handling, cryptographic operations, and background tasks still needed to be designed efficiently.

6.2 Symbian OS

Symbian was designed for constrained mobile hardware and was known for efficient use of memory, CPU time, and battery power. Its multitasking and native APIs allowed developers to build responsive applications, but developers had to manage resources carefully. Poorly designed applications could still drain the battery, leak memory, or destabilize a device.

Resource factor	BlackBerry OS	Symbian OS	Assessment
Memory and CPU	Optimized for mobile messaging, but capacity depends on the model and generation.	Designed to operate on constrained mobile hardware; native code can be efficient but requires careful engineering.	Symbian can be very efficient; model-specific benchmarks are needed.
Battery usage	Push-oriented messaging	Efficient system design can	Architecture and

	may reduce repeated polling when supported by the service.	conserve power, but network polling and background tasks affect battery life.	application behaviour matter more than the OS label alone.
Multitasking	Supports background communication and device-dependent background services.	Supports multitasking and system services on relevant versions.	Both can run messaging tasks; test concurrency and notification behaviour.
Reliability under load	Enterprise-managed workflows may be predictable, but old hardware and software can limit performance.	Efficient when applications respect platform resource rules; device variation complicates consistency.	Measure delivery latency, memory use, and battery drain on actual devices.
Attachment handling	Large files can strain storage, memory, and network capacity on older models.	Same limitation; streaming or size limits may be needed.	Use attachment-size limits, compression where appropriate, and bounded local storage.

7. Architecture Diagrams

Diagram 1: Traditional BlackBerry Enterprise Messaging Flow

[Employee BlackBerry Smartphone]

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[BlackBerry Network / Secure Transport]

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[BlackBerry Enterprise Server (BES)]

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[Corporate Mail / Messaging Server]

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[Enterprise Administrator and IT Policies]

Explanation: In a traditional managed deployment, the device communicates through the BlackBerry messaging infrastructure and enterprise server components. The corporate server and administrator

apply the organization's messaging and device policies. Exact routing and components vary by BlackBerry generation and deployment.

Diagram 2: Generic Symbian Messaging Application Flow

[Employee Symbian Smartphone]

⇅

[Messaging Application / Runtime]

⇅

[Symbian Networking APIs]

⇅

[Cellular Data / Wi-Fi, if supported]

⇅

[Corporate Messaging Server]

⇅

[Authentication + TLS/VPN + Server Policy]

Explanation: A Symbian application typically uses the networking facilities available on the handset to communicate with a server. Secure transport, authentication, message storage, and enterprise policy must be implemented or supplied by the selected application and infrastructure.

Diagram 3: Simplified Security Layers

Layer	BlackBerry OS enterprise approach	Symbian approach
User identity	Device/account authentication and enterprise account controls.	Device/application authentication according to the selected solution.
Application	Application restrictions and enterprise policy where supported.	Application signing, capabilities, and process isolation on supported versions.
Data at rest	Device and enterprise data-protection features depend on generation/configuration.	Data caging and protected storage; encryption depends on device/configuration.
Data in transit	Enterprise messaging transport and configured cryptographic protections.	Secure application protocols such as correctly configured TLS/VPN, if supported.
Administration	Centralized BES-era management in compatible deployments.	Device/vendor management tools or third-party solutions, depending on

		fleet.

8. Comparative Analysis

For secure enterprise messaging, the decisive factor is not simply whether an operating system has security features. The organization also needs a complete operational system: secure transport, user authentication, controlled application installation, central policy enforcement, device-loss response, and compatibility with corporate servers.

BlackBerry OS has the stronger fit for a historical enterprise messaging scenario because its ecosystem was built around managed messaging and administrator control. A BlackBerry Enterprise Server deployment could provide a centralized path between the corporate messaging environment and enrolled devices, making policy enforcement and message delivery easier to manage consistently.

Symbian should not be described as inherently insecure. Its capability-based model and data caging were meaningful design protections, and its efficient resource use could be beneficial on older smartphones. Nevertheless, an organization may need to combine these OS-level mechanisms with a secure messaging application, secure transport, and separate device-management tools. Variation among Symbian devices also increases the testing and administration burden.

The conclusion changes if the existing BlackBerry devices or server infrastructure are no longer supported. A well-designed Symbian solution with strong authentication, correctly configured encryption, minimal local storage, and reliable management could be safer than an unsupported BlackBerry deployment. Security must be assessed against the exact model, OS release, patch level, and server configuration.

9. Decision Matrix

Evaluation criterion	BlackBerry OS	Symbian OS	Preferred for this scenario
Enterprise security integration	Strong in compatible traditional BES deployments.	Available, but often more dependent on application and device configuration.	BlackBerry OS
Application-level security	Depends on generation, app controls, and deployment.	Capabilities, signing, and data caging on supported	Context-dependent

		versions.	
Messaging/network integration	Historically designed around enterprise messaging and push delivery.	Flexible network APIs; messaging depends on client/server implementation.	BlackBerry OS
Runtime flexibility	Platform-specific legacy runtime and APIs.	Native and Java ME options on supported versions.	Symbian for flexibility; neither for easy modern maintenance
Resource efficiency	Designed for mobile communication and low-power operation.	Known for constrained-device efficiency; application quality remains decisive.	Symbian, potentially
Centralized administration	Strong with compatible enterprise infrastructure.	Depends on available management tools and handset vendor.	BlackBerry OS
Current deployment viability	Must verify server, certificates, support, and device compatibility.	Must verify OS support, security patches, and compatibility.	Neither without a support review

10. Recommendation

BlackBerry OS is the better choice for the stated scenario if the organization already owns compatible legacy BlackBerry smartphones and can operate a supported, correctly configured enterprise messaging environment. Its historical strengths in enterprise messaging, centralized administration, and policy enforcement align closely with the requirement to protect employee communications.

However, this is a conditional recommendation, not a claim that every BlackBerry device is secure today. Before deployment, the organization should verify that the exact devices and server components are still supported, apply all available security updates, require strong authentication, enforce encrypted transport, minimize sensitive data stored locally, and test remote-lock/wipe procedures. If the legacy infrastructure is unsupported or cannot meet current security requirements, the organization should migrate to a currently supported platform rather than rely on either legacy OS for sensitive business messaging.

11. Conclusion

BlackBerry OS and Symbian both introduced important mobile security and efficiency features, but they addressed enterprise messaging differently. BlackBerry's traditional ecosystem offered a more integrated approach to corporate messaging and centralized administration. Symbian provided strong platform mechanisms such as application signing, capabilities, and data caging, together with efficient resource management and flexible application development.

Therefore, for a legacy enterprise messaging deployment, BlackBerry OS is generally the better fit because its messaging and administration model aligns more directly with the organization's needs. The final decision must still depend on the actual device generation, available security updates, network configuration, and ongoing support. Unsupported legacy devices should not be used for sensitive communications without a formal risk assessment.